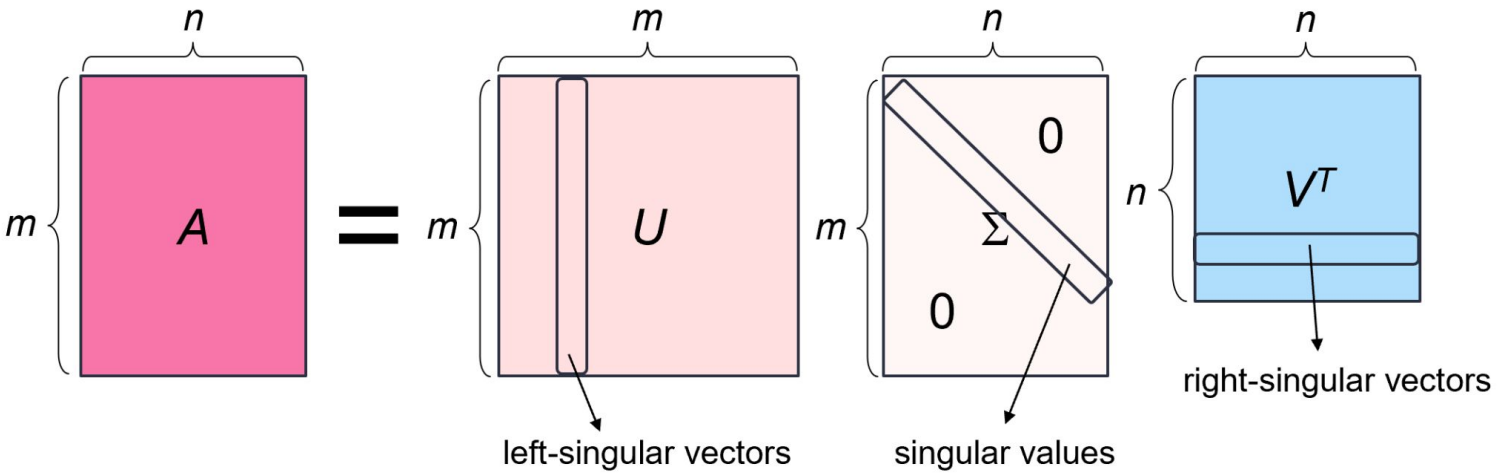
Tensor Decomposition

SVD decomposition



Original frame

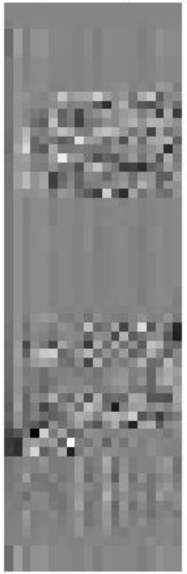


Resized frame

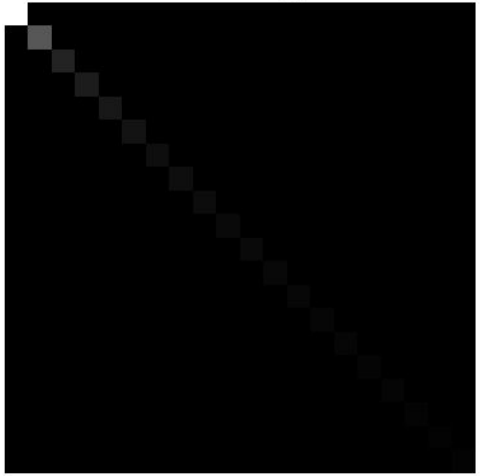


SVD decomposition of an image

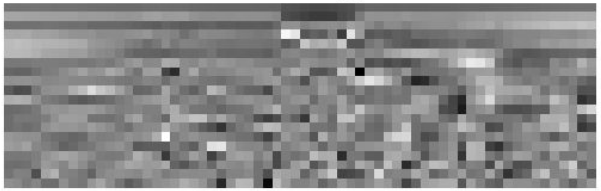
Frame 1 right singular vectors



Frame 1 SVD rank 5 approximation



Frame 1 SVD rank 5 approximation

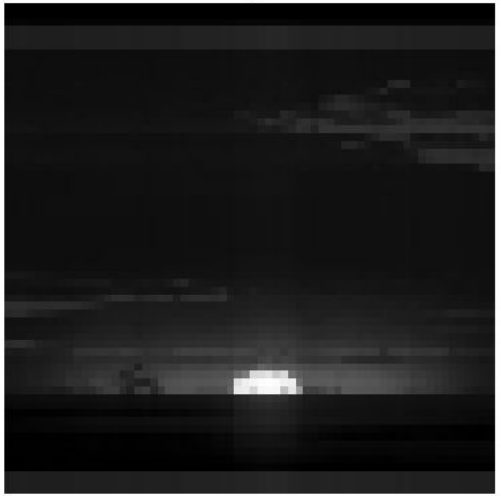


Changing feature of image in Tucker through eigenvalues changed

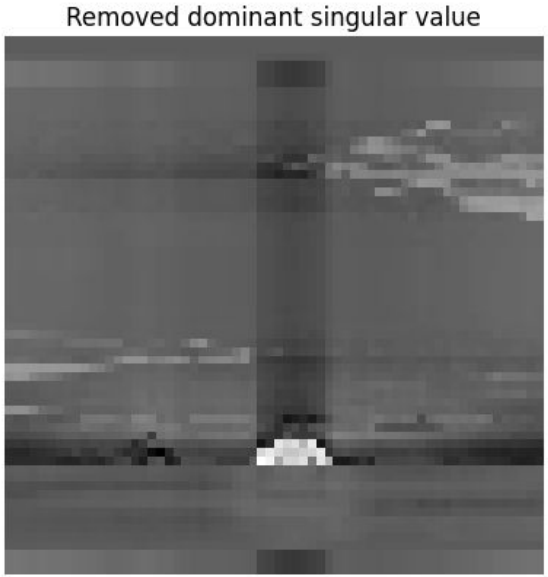
Frame 1 Reconstructed from SVD rank 5 approximation



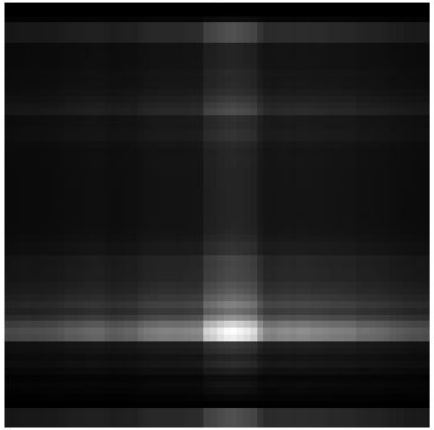
Boosted singular values



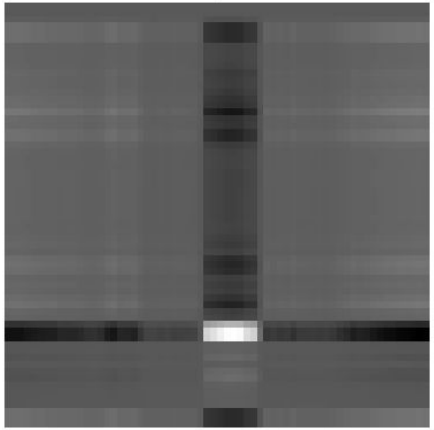
Removing dominant features in Tucker



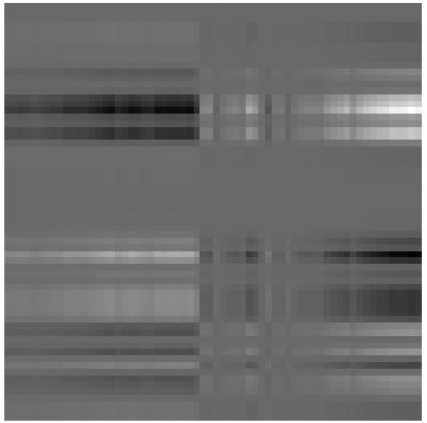
SVD Component 1



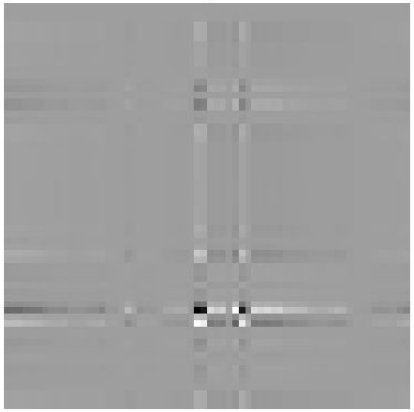
SVD Component 2



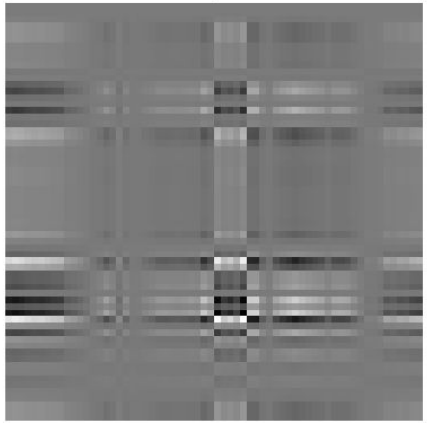
SVD Component 3



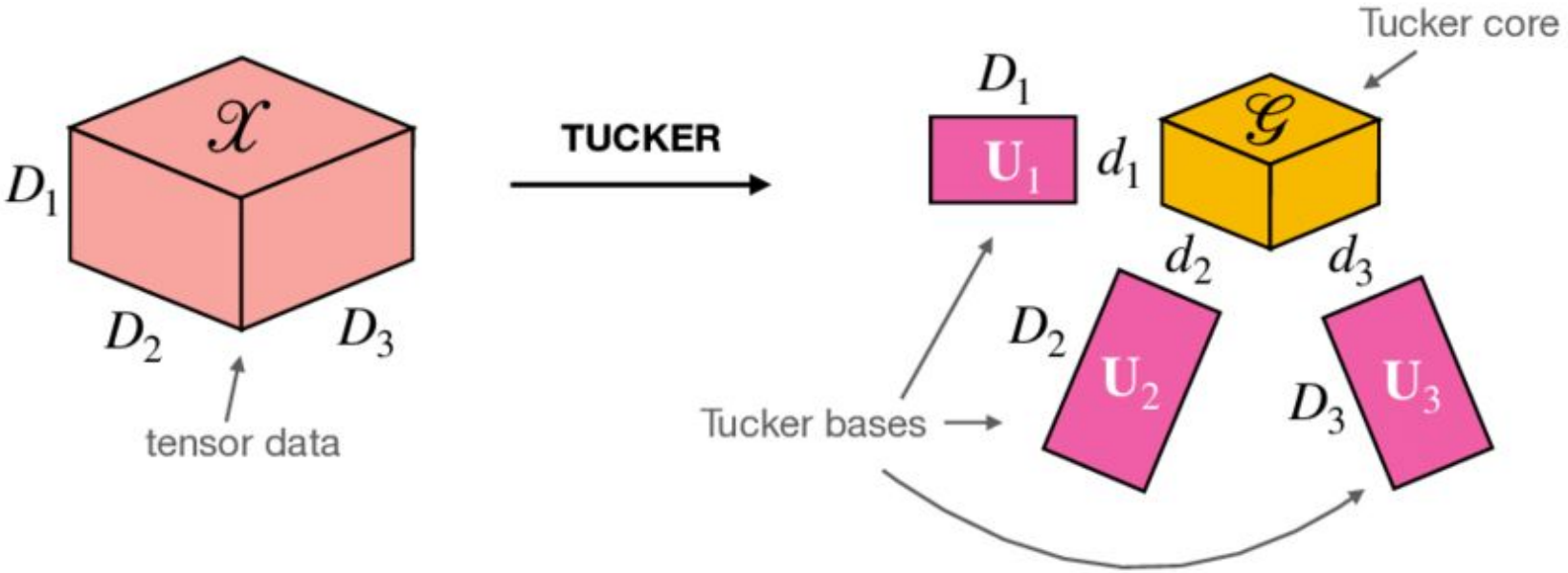
SVD Component 4



SVD Component 5



Tucker Decomposition

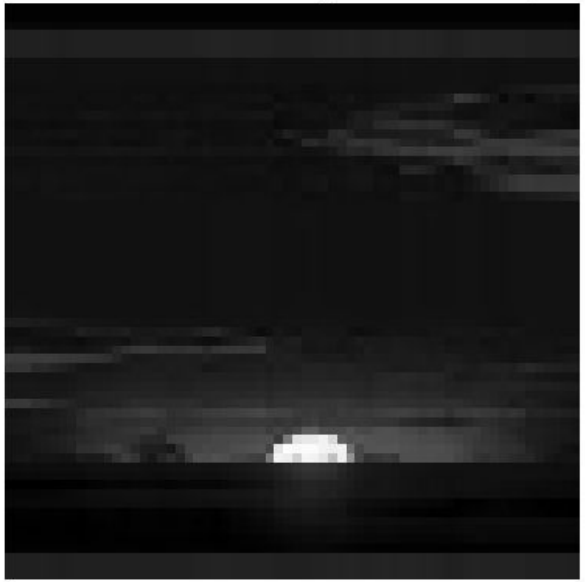


Tucker and Original frame comparision

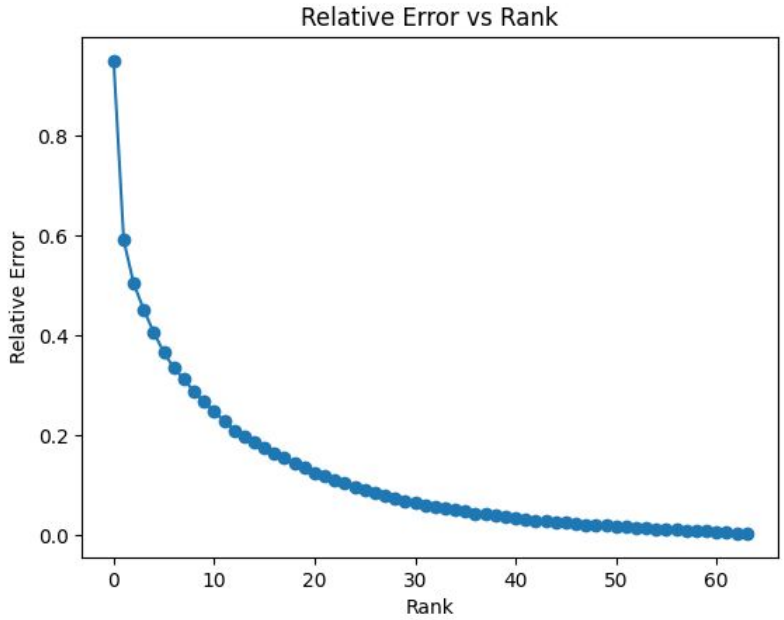
Original frame

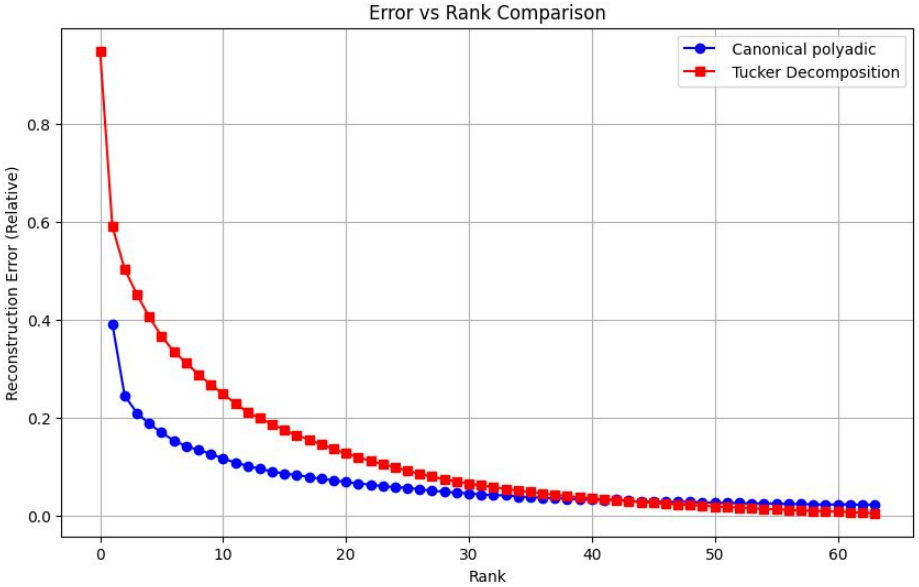


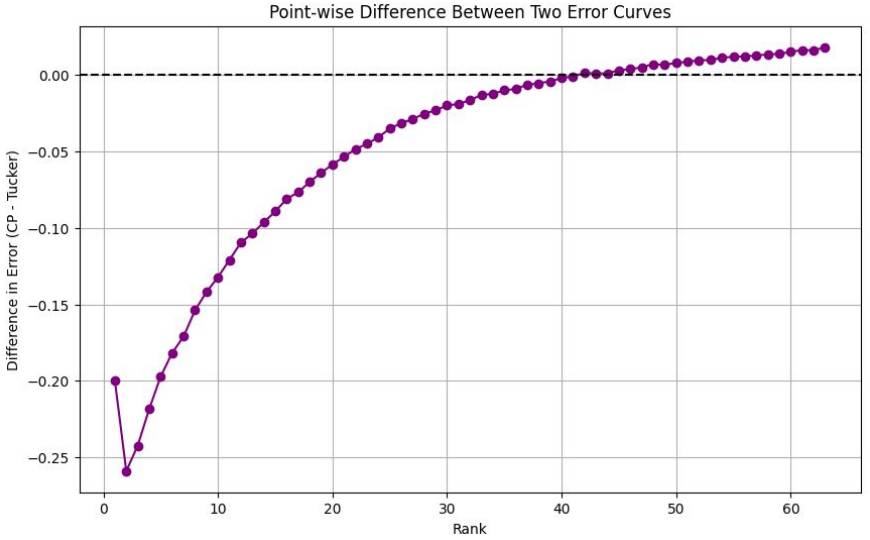
Reconstructed Frame 1 using Tucker Decomposition



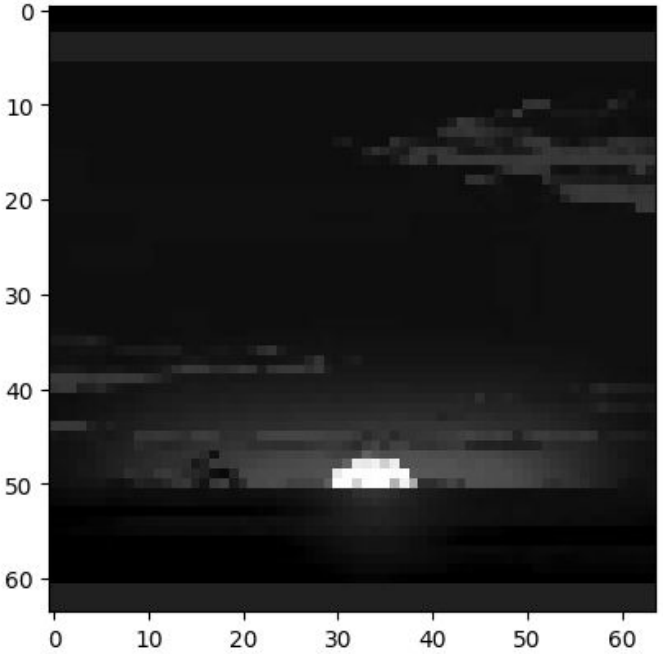
Tucker Decomposition Error vs Rank



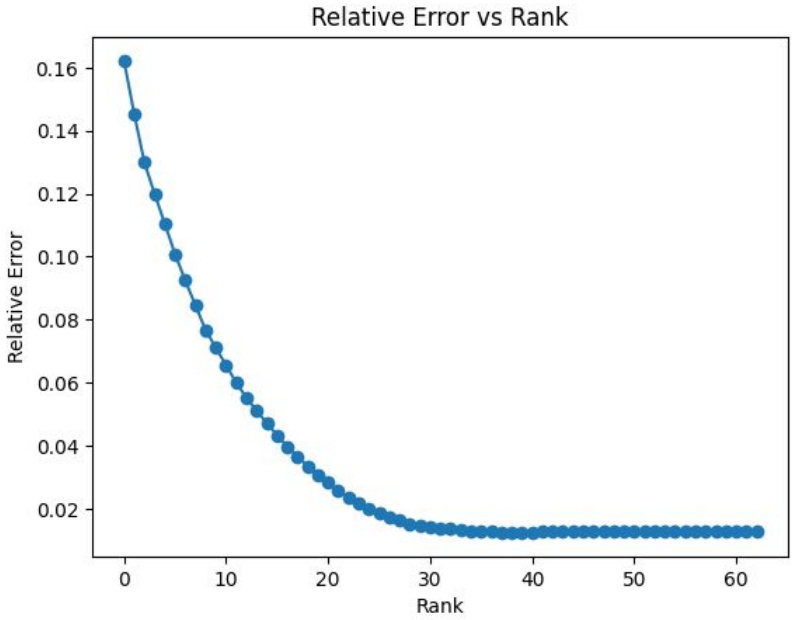

Error vs Rank Comparison
Canonical polyadic
Tucker Decomposition
Reconstruction Error (Relative)
Rank
0.0
0.2
0.4
0.6
0.8
0
10
20
30
40
50
60


Point-wise Difference Between Two Error Curves
Difference in Error (CP - Tucker)
Rank
0.00
−0.05
−0.10
−0.15
−0.20
−0.25
0
10
20
30
40
50
60

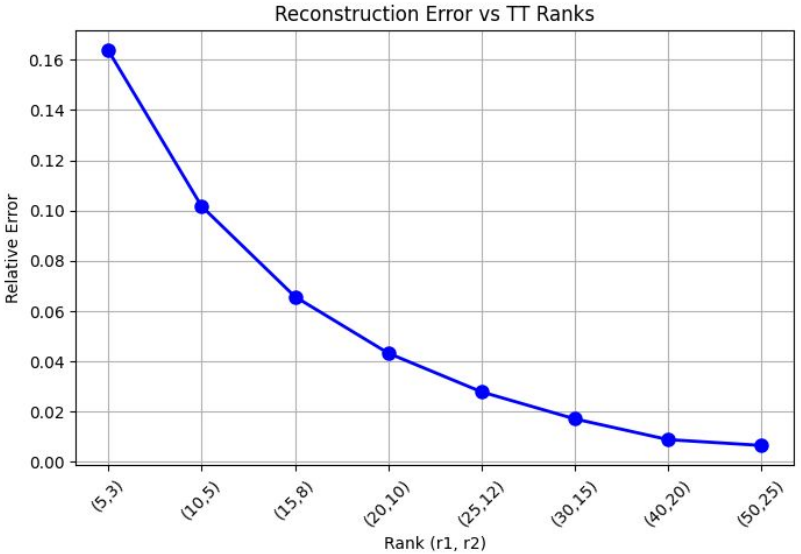
Tensor Train decomposition

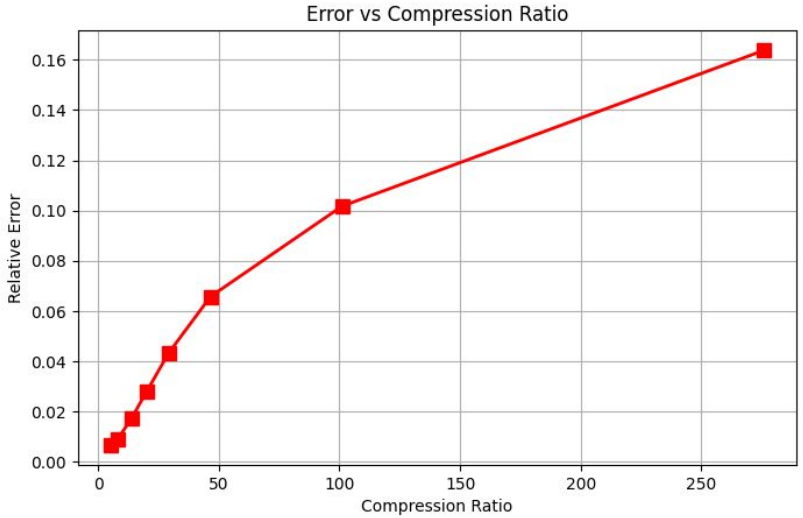


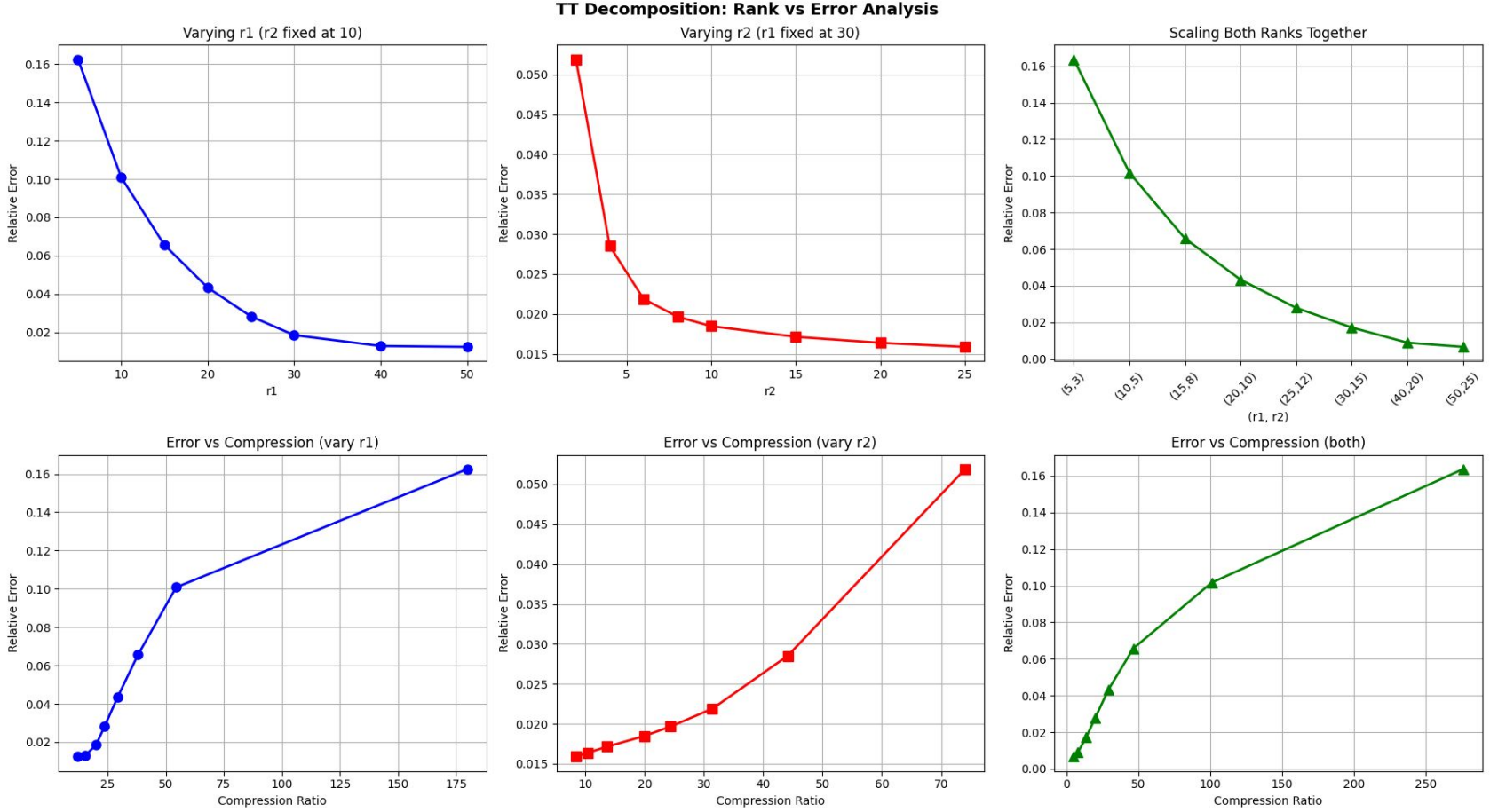
Given R2 stays fixed whilst R1 changes from rank 1 to 63



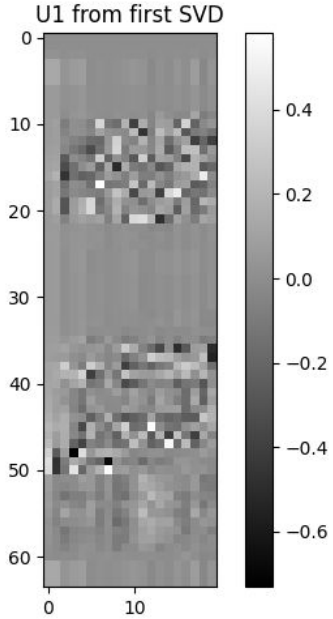
Tensor Train Rank vs Error

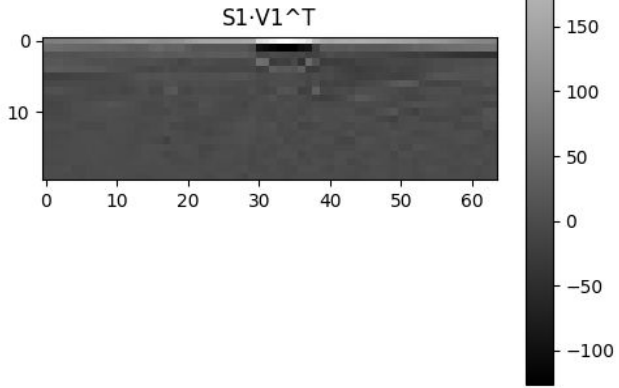


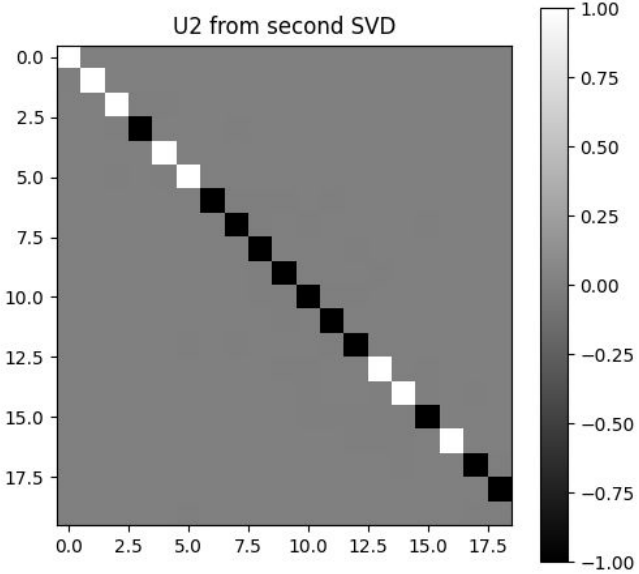


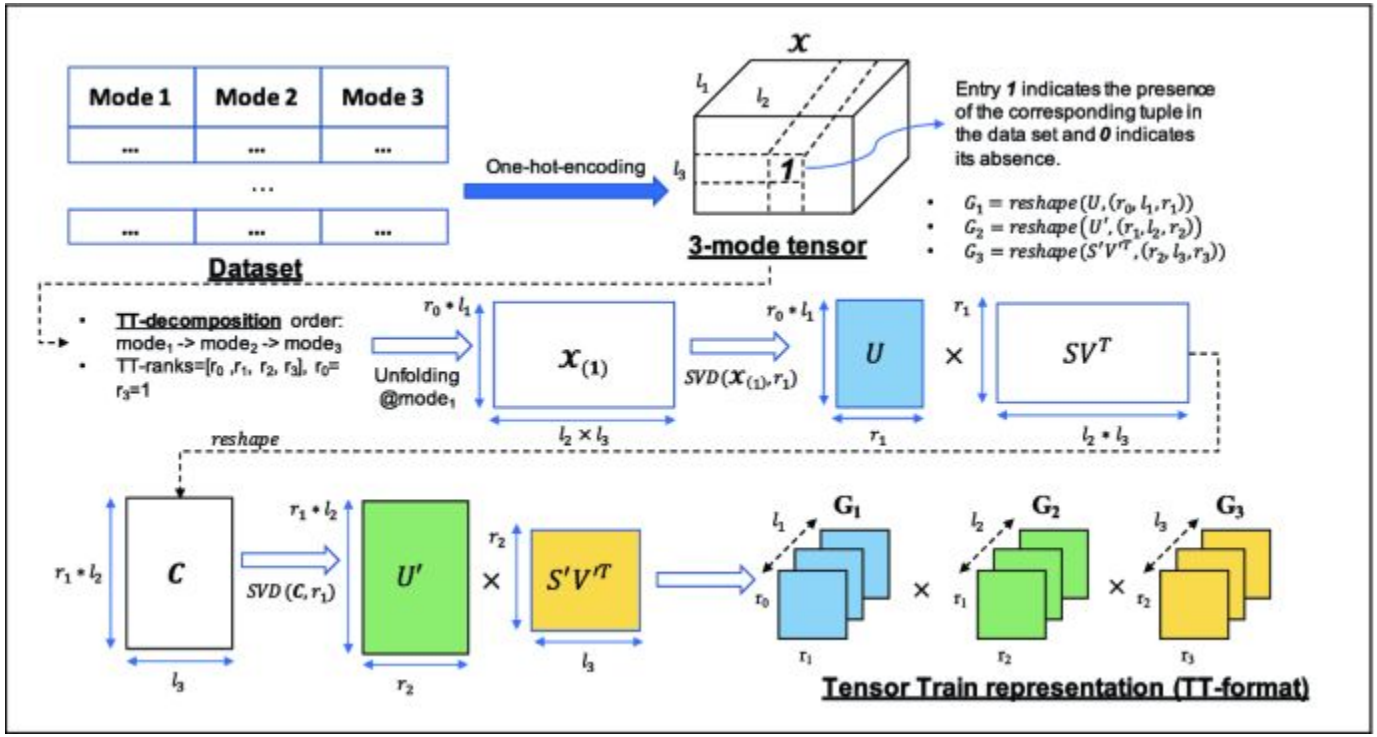

TT Decomposition: Rank vs Error Analysis
Varying r1 (r2 fixed at 10)
Relative Error
r1
Varying r2 (r1 fixed at 30)
Relative Error
r2
Scaling Both Ranks Together
Relative Error
(5,3)
(10,5)
(15,8)
(20,10)
(25,12)
(30,15)
(40,20)
(50,25)
(r1, r2)
Error vs Compression (vary r1)
Relative Error
Compression Ratio
Error vs Compression (vary r2)
Relative Error
Compression Ratio
Error vs Compression (both)
Relative Error
Compression Ratio

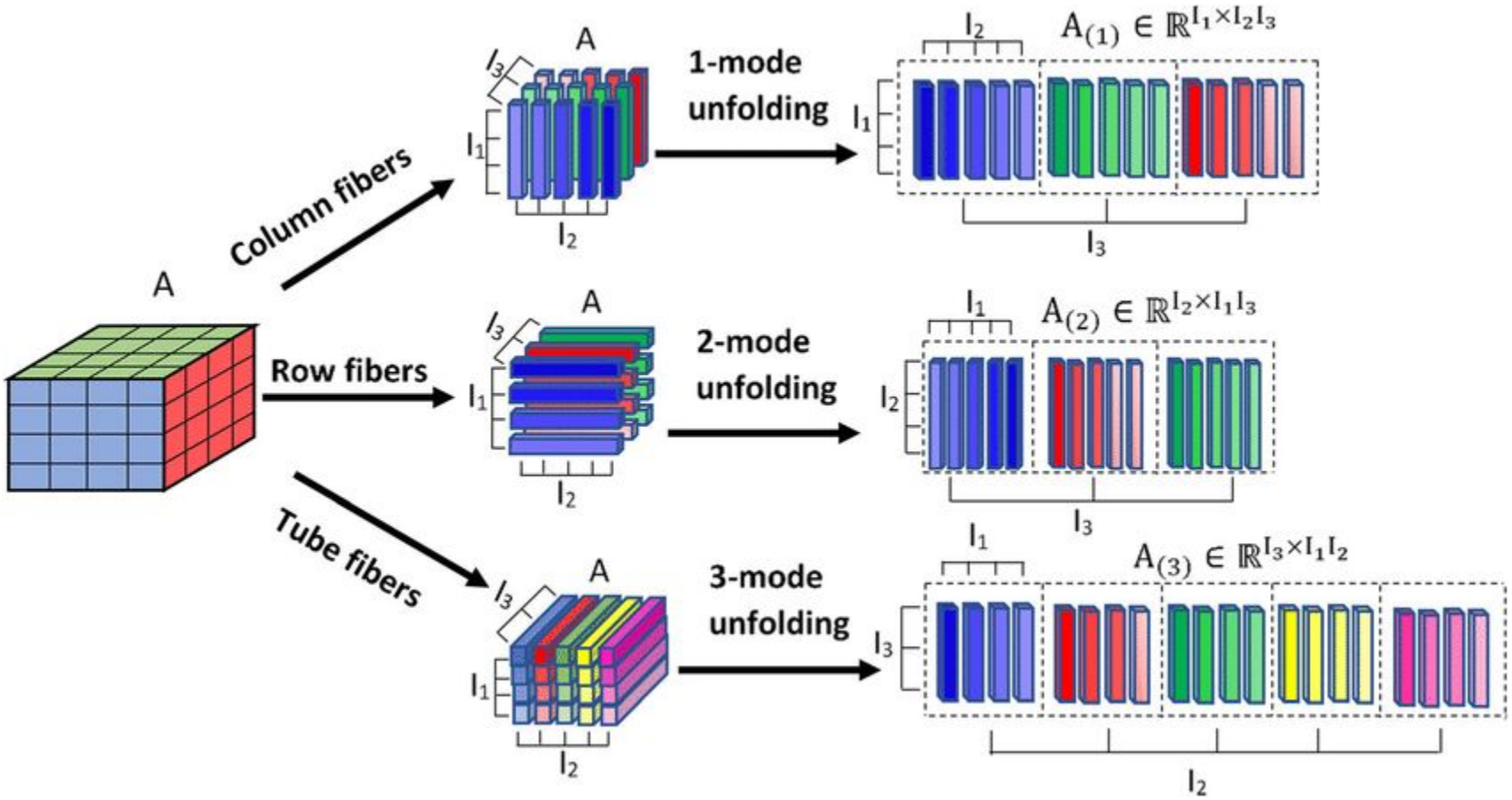
The different SVD transformation and what they are








Mode 1
Mode 2
Mode 3
Dataset
One-hot-encoding
3-mode tensor
Entry 1 indicates the presence of the corresponding tuple in the data set and 0 indicates its absence.
$G_1 = reshape(U, (r_0, l_1, r_1))$
$G_2 = reshape(U', (r_1, l_2, r_2))$
$G_3 = reshape(S'V'^T, (r_2, l_3, r_3))$
TT-decomposition order: mode1 -> mode2 -> mode3
TT-ranks=[r0 ,r1, r2, r3], r0= r3=1
Unfolding @mode1
$SVD(\mathcal{X}_{(1)}, r_1)$
reshape
$SVD(C, r_1)$
Tensor Train representation (TT-format)


A
Column fibers
Row fibers
Tube fibers
A
I3
I1
I2
1-mode
unfolding
I2
A(1) ∈ ℝ^I1×I2I3
I1
I3
A
I3
I1
I2
2-mode
unfolding
I1
A(2) ∈ ℝ^I2×I1I3
I2
I3
A
I3
I1
I2
3-mode
unfolding
I1
A(3) ∈ ℝ^I3×I1I2
I3
I2

SVD and how left singular is in row space

For a tensor:

$$\mathcal{X} \in \mathbb{R}^{I \times J \times K}$$

Mode-1 unfolding gives:

$$X_{(1)} \in \mathbb{R}^{I \times (JK)}$$

Now do SVD:

$$X_{(1)} = U^{(1)} \Sigma^{(1)} V^{(1)T}$$

Look at dimensions:

- $U^{(1)} \in \mathbb{R}^{I \times I}$

Residual:

$$R_1 = \Sigma_1 V_1^T \in \mathbb{R}^{r_1 \times (I_2 I_3)}$$

Now we reshape to bring **mode-2 into the row dimension**:

$$R_1 \to R_1^{(2)} \in \mathbb{R}^{(r_1 I_2) \times I_3}$$

- Rows now = previous rank r_1 × mode-2 dimension I_2
- Columns = mode-3 dimension I_3

Mode-1 fibers ⟶ first decomposition

Mode-2 fibers ⟶ embedded in the rows of the reshaped residual

Mode-3 fibers ⟶ remaining columns

Tensor Train decomposition and how modes behaves with each other

r1r_1r1 = **number of intermediate states connecting mode-1 to mode-2**
r2r_2r2 = **number of intermediate states connecting mode-2 to mode-3**

i1 – r1– > i2 – r2 --> i3

r1 = “how much mode-1 communicates with mode-2”
r2 = “how much mode-2 communicates with mode-3”

Effects of changing the linking variable

Rank scenario	Where complexity is	Effect
Large r1, small r2	Mode-1 & Mode-2	Early modes dominate, last mode simple
Small r1, large r2	Mode-2 & Mode-3	Late modes dominate, first mode simple
Large r1, large r2	All modes	Tensor highly expressive, less compression
Small r1, small r2	All modes	Strong compression, can’t capture complex correlations

Curse of dimensionality

Let’s say:

- n=10n = 10n=10
- d=20d = 20d=20
- TT rank r=10r = 10r=10

Full tensor:

10^{20} =

Tensor Train:

20×10×10= 20,000

Tensor Decomposition
O(d n r^2)
Strongly mitigated
Efficient for structured, compressible tensors

Future Research Prospect